



OPTRONIC SENSORS

Fundamentals And Types



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Optronic sensors constitute the heart of a variety of systems ranging from simple gadgets like light meters to the most complex of military systems like precision-guided munitions, laser rangefinders, target trackers, remote sensing systems, navigation sensors, sniper and explosive detectors, fibre-optic and laser-based communication systems, night vision devices, Lidar and spectroscopic sensors. While individual photosensors such as PIN photodiodes and avalanche photodiodes find applications in military optronic systems including laser rangefinders and target designators, Lidar sensors and navigation sensors, sensor arrays such as complementary metal-oxide semiconductor (CMOS), charge-coupled device and avalanche photodiodes are at the core of imaging sensor systems including Ladar sensors, night vision devices, laser and imaging infrared seekers.

Types of photosensors

Photosensors are classified into two major categories: Photoelectric and thermal. Photoelectric sensors are further of two types: Devices that depend on the external photo effect for their operation and devices that make use of some kind of internal photo effect.

Photo-emissive sensors are based on external photo effect. Common photo-emissive sensors include non-imaging sensors such as vacuum photo cells and photomultiplier tubes, and imaging sensors such as image intensifier tubes.

Photoconductors and junction-type photosensors use internal photo effect. Photoconductors are bulk semiconductor devices whose resistance decreases with increase in incident light intensity. These are also known by the name of photoresistors, light-dependent resistors and photocells.

Junction-type photosensors are further amplifying and non-amplifying type. Amplifying-type junction photosensors include phototransistors, photothyristors and photo-field-effect transistors (FETs). Non-amplifying types of junction photosensors include photodiodes, solar cells, CMOS

and charge-coupled devices. CMOS and charge-coupled device are imaging sensors. The category of thermal sensors includes thermocouple (or thermopile) type sensors, bolometric sensors and pyroelectric sensors.

Thermal sensors absorb incident radiation and operate on the resulting temperature rise, whereas photoelectric sensors are based on quantum effect. Compared to photoelectric sensors, thermal sensors are sluggish in their response to the incident radiation. However, these offer a much wider operational wavelength band than photoelectric sensors.

Characteristic parameters

Major parameters used to characterise the performance of photosensors include responsivity, noise equivalent power (NEP), sensitivity (usually measured as detectivity or D^*), quantum efficiency, response time and noise.

Responsivity is the ratio of electrical output to radiant light input determined in the linear region of response. It is measured in amperes per watt (A/W) or V/W if the photosensor produces a voltage output rather than a current output. Responsivity is a function of the incident radiation's wavelength and band-gap energy. Spectral response is a related parameter. It is a curve that shows variation of responsivity as a function of wavelength.

Most photoelectric sensors have a narrow spectral response, whereas most thermal sensors have a wide spectral response. As an example, spectral responses of silicon, germanium, indium-gallium-arsenide photodiodes are in the range of 200-1100nm, 500-1900nm and 700-1700nm, respectively, in comparison to thermistors' 0.5-10µm.

Silicon photodiodes exhibit a response from ultraviolet through visible and into near infrared part of the spectrum. With silicon having band-gap energy of 1.12eV at room temperature, its spectral response peaks in the near infrared region between 800nm and 950nm. Peak responsivity figures for silicon PIN photodiodes are in

Understanding optronic sensors and related sensor systems is essential to understanding design and operation of a large number of defence systems. Hence this six-part article series starts with an overview of these sensors